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Individual Assignment 1

Simple Shopping Bill

1. Case Narrative

A customer buys several items of the same type and price. The cashier needs to calculate the total amount that the customer must pay. The calculation is simple: multiply the price of one item by the number of items purchased.

This algorithm does not include a discount, tax, or additional charge. The price must be zero or greater, and the quantity must be a positive whole number.

2. Input, Process, and Output

Part	Description
Input	price: price of one item; quantity: number of items purchased
Process	Calculate total = price * quantity
Output	total: the amount that the customer must pay

3. Narrative Representation

1. Start.
2. Read the price of one item.
3. Read the number of items purchased.
4. Multiply the price by the quantity to calculate the total.
5. Display the total amount to pay.
6. Stop.

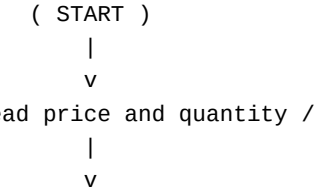
4. Pseudocode Representation

```
ALGORITHM SimpleShoppingBill

    DECLARE price, total AS number
    DECLARE quantity AS integer

    START
        READ price
        READ quantity
        total <- price * quantity
        DISPLAY total
    STOP

END ALGORITHM
```

5. Flowchart Representation

The flowchart uses the following standard symbols:

â¢ Oval: start or stop
â¢ Parallelogram: input or output
â¢ Rectangle: process
â¢ Arrow: direction of flow



```
          [ total <- price * quantity ]
                  |
                  v
            / Display total /
                  |
                  v
               ( STOP )
```

The same flow is represented below in Mermaid notation for a graphical rendering:

```
flowchart TD
    A([START]) --> B[/Read price and quantity/]
    B --> C[total <- price * quantity]
    C --> D[/Display total/]
    D --> E([STOP])
```

6. Test with Sample Data

Test Case

Variable	Value
price	Rp 25,000
quantity	4

Trace of Execution

1. The algorithm reads price = Rp 25,000.
2. The algorithm reads quantity = 4.
3. It calculates:

   ‘‘‘text
   total = price * quantity
         = 25,000 * 4
         = 100,000
   ‘‘‘

4. It displays Rp 100,000.

Result

```
Input:
  price = Rp 25,000
  quantity = 4
```

```
Output:
  total = Rp 100,000
```

The result is correct because four items priced at Rp 25,000 each cost Rp 100,000 in total.

7. Conclusion

The solution is a finite sequence of ordered steps. The narrative explains the algorithm in everyday language, the pseudocode expresses the same logic using structured text, and the flowchart expresses it using connected symbols. All three representations preserve the same input, calculation, and output. The trace with sample data confirms that the algorithm produces the expected result.